

Ambreeesh P

AI - Assisted Drone for Disaster Management, which is a prototype, is built around a Raspberry pi 5

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Abstract

Disasters Floods, earthquakes, fires, and industrial accidents are some of the disasters that often provide unsafe and inaccessible conditions in which quick rescue is difficult. Under this case, it is vital to detect the affected individuals early enough to minimize loss of life. This paper dwells on the Design and Development of an AI-Assisted Drone System to help manage disasters in the natural situation.

The suggested system is interpreted as a quadcopter platform with Pixhawk flight controller board and processing unit on the platform based on Raspberry Pi. It has a camera on the drone capturing real-time video and photographs on the flight, which is analyzed in real-time by AI and CV methods. OpenCV is applied to perform image processing and annotations, whereas human detection is conducted with the help of the YOLOv8 model. The system can detect human presence and recognize any distressed related gesture with the help of a pose estimation model, like a raised hand, and it would need urgent help.

Raspberry PI communicates with the flight controller to maintain a stable flight and handles data properly, the system was able in BOT API to recognize people and produce the appropriate alerts in a timely manner and provide visual information as well as GPS positioning in the case of monitoring when on the ground.

This project points out the idea of the AI-powered drones as the future of disaster response to a better situational awareness, a decrease in human risk, and a quicker nature of the rescue response.

1. Introduction

Every year natural and manmade catastrophes like floods, cyclones, earthquakes, disasters, and fires and industrial mishaps take place in Recent times. The consequences of these disasters result in huge loss of human life and property. In such cases the operations of rescuing will also be challenging due to the geographical areas and can be hazardous to Find rescued individuals in disaster afflicted places. The safety and a delay, which may lead to the human life risk, is one of the greatest issues faced by rescue teams.

Drones have found their use in disaster management since they are able to travel to hazardous and inaccessible locations and give real-time images and videos. This assists authorities to know more about the situation. Nevertheless, drone systems used by majority of people need constant human supervision and this is time consuming and can cause errors.

To counter this drawback, smart systems are necessary that will be able to process visual information by automatic means and implement the Challenges, the development and design of an AI Assisted Drone For Disaster Management is a prototype that will be deployed to help save lives through more efficient identification of issues of distress and human beings.



Figure1.1 Detection of a Distress Person

Design and Development of AI – Assisted Drone for Disaster Management

1

1.1 Visual Challenges in Disaster Scenarios

Disaster The conditions of a disaster are normally unpredictable. There are poor conditions of lighting thus making the analysis hard to visualize. The images taken in case of disasters can be blurred because of the movements of the drones or because of low visibility and then they are surrounded by water and therefore can be detected by human beings with more difficulty. Such poor-quality images are one of the primary research issues which involve identifying humans.

Under such circumstances, image processing approaches are liable to failure.

Image recognition systems should be in a position of dealing with noise, and varying lighting conditions.

One of the significant research issues in the field of disaster management application is the development of such robust systems.

This project examines that image recognition methods using AI can be used to identify humans even on challenging environments when they are considered

difficult to view. Through concentration on real time image analysis using Visual Enhancement techniques the system will enhance reliability of the human detection in times of disaster.

The image-based monitoring technique is very important in detecting the presence of drones near airports and within runways.

1.2 Significance of Drones in Image-Based Monitoring

Image-based monitoring method is quite essential in identifying the existence of drones in around airports and runways.

In the event of a disaster, the use of drone is of great significance to the present-day disaster monitoring. They offer aerial images which cannot be viewed through the ground-based techniques. A drone has the capability to serve a large area in a limited time period and give real time images of the point of disaster, This assists the authorities to have visuals of the damages and organize rescue missions more efficiently. This data is then too hard to analyse manually and is life threatening. This brings out the necessity of automated systems of monitoring through images. Together with image recognition methods, it is also possible to develop systems capable of capturing images and, at the same time, analyse them intelligently. This project aims at enhancing the practicality of monitoring using drones through image recognition.

1.3 Motivation and Need for the project

long rescue operations usually arranged by local parties or NDMA, usually put their lives in danger when undertaking rescue operations,

Human operators need to pay full attention to the video feeds of the drones and this makes them fatigued and they can overlook critical information.

Automatically performing an analysis of images and recognizing humans is done by image recognition techniques. With AI, drones can identify individuals on the spot,

and emphasize on valuable information on sight. This minimizes manpower required by the human operators and increases the speed of response.

In the project, humans can be detected by their image recognition, and basic gestures that can also report distress are identified. The system takes the live video frames and gives useful information to the rescue teams. This proves that image recognition is significant in enhancing disaster response mechanisms.

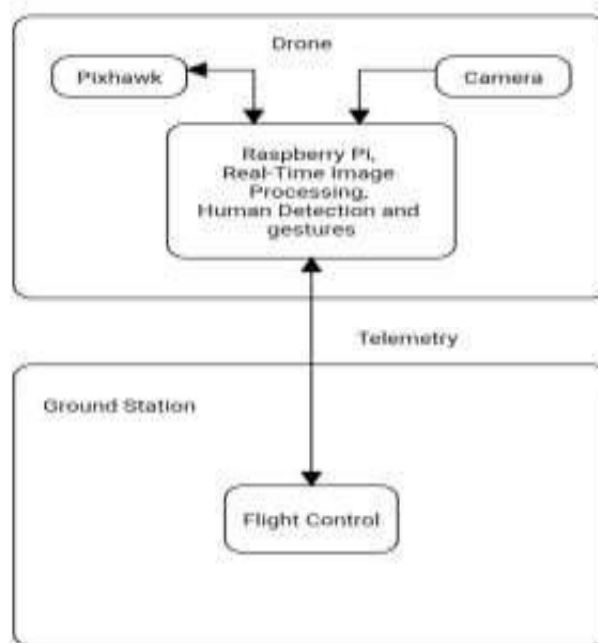


Figure 1.2 Block diagram of the system

1.4 Future prospects and potential enhancements

In a state such as India, where natural calamities are common, and they affect huge numbers of people, vision bio type systems may be crucial in enhancing effectiveness and safety during rescue operations. To some cases the communication aspects might be compromised, so the communication of the drone to the ground station should be more

improved and enhanced, The likes of fish camera lens and thermal cameras among other camera lens might be used to see a better result of the situation i.e. a person trapped under a slab or in a building the use of thermal cameras might help.

2. Background Theory

By Through the study of the previous work, mapping of how similar problems have been approached in the past and the methods used, visual analysis and the research of the limitations of the established systems is set, like the manual detection of a human body will occupy lots of labour and time wasting which in such situations is very essential and such research has high percentile of error.

Various researches have focused on UAV based detections, including a basic examination of conditions to self-flying drone, although in most cases there is the challenge of drone control, image enhancements and communications, low range and certain challenges few.

2.1 Foundational Studies in UAV-based Human Detection.

Some of the early approaches to human monitoring with the use of UAVs relied on more traditional computer vision protocols, Gradients were coined with established classifiers such as

Local Binary Patterns and FPDW Support Vector Machines to detect humans in aerial images Combination of measures such as HOG with SVM, Local Binary Patterns and FPDW had fair robustness to variation, though their performance critically relied on the selection and tuning of features.

At the same time, there were drones used in photography and general surveillance. Later experiments commenced on more specific uses in the context of human detection and interaction. it is explicitly indicated in these studies that the drones with cameras can observe and analyse human presence in aerial perspectives. Nevertheless, the majority of these early works were aimed only at human detection as object and did not regard the analysis of the body posture or meaningful gesture acknowledgements.

2.2 Principles and Assumptions

The aerial human gesture detection systems rely on a number of geometric as well as operational assumptions. In other words, the size of the human beings in the image is defined by the altitude of the drone and the resolution of its camera. Practically, the bigger the resolution of the camera, the higher the flight altitude, provided that we indulge too much in high flight or have a low-resolution camera, the pixels are small which is detrimental to detecting human beings.

Algorithms.

On the algorithmic side, a large number of models are based on the ability to use large training datasets of labelled examples. In reality, aerial vision systems cannot work without people of interest being seen either obliquely or above, and without the model having been trained or fined-tuned on visually similar images.

The major assumptions are:

- Drones' stability: it should be as stable as possible so that the camera does not get blurry products, its stability depends on several factors such as Central Gravity(CG) and aerodynamic disturbance.
- loiter mode: the drone should be stationary when an operation is in process, like the period of time the data-analytics may be perfect and it may be utilized in mapping an rescue strategy.
- environmental conditions: the drone should not be affected by environmental factors like the wind, turbulence etc.
- low light improvement method: the low light situation renders the image of the human being very hazy and unattainable thus improving the +video is an added advantage in the given project.
- High resolution: makes the detection from a far visual point easy, as it can clearly detect and analyse the key points in the captured frames.

2.3Key Definitions

- UAV (drone): This is an unmanned flying vehicle which has cameras/sensors.
- Object Detection: The object recognition and classification problem in an image (persons in this case).
- Human Pose Estimation: Detecting important joints locations (e.g. elbows, knees) of a person. Usually in the form of a 2D skeleton or heatmaps.
- Gesture Recognition: Categorizing a human body pose or movement into a redefined set Gestures here are restricted to a set of rescue gestures such as waving arms to draw attention.
- Real-Time System: This is a system that can autonomously handle every video frame (or a video stream) at a sufficiently fast frame rate (typically [?]30 fps) to be interactive.

- **System Throughput:** The frames per second (FPS) value that is obtained points to viability in real time. On board UAV systems are usually aimed at 10- 30 FPS based on equipment.
- **Drone stability:** Stability in this context entails the fact that despite the effects on the environment and regardless of the weakness in the structural parts the drone remains at a stable position.
- **loiter mode:** Loiter Mode is an automatic mode, and it tries to keep the current location, heading and altitude.
- **environmental conditions** There are a few major conditions that influence drone surveying, which include temperature, wind, humidity, rain, and clouds.
- **image enhancement on low light:** low-light image improving process is capable of raising image brightness and more accurately restoring colour and detail information.
- **High quality:** High quality drones refer to those drones that have the capacity to take detailed imagery, which are required in professional aerial photographing, inspection, and surveying.

2.4 Merits, Demerits and Applicability.

Merits: Aerial vision gives various situational awarenesses. Drones are able to reach isolated or unsafe areas and survey a greater area in a short period and provide an aerial perspective unattainable to ground troops. They are able to travel tens of square kilometres per hour, and thus can search victims fast. Detection based on vision is non-invasive: the wearable sensors do not make the victim carry an item unlike vision-based detection. Shopping Cart 2 is also a supplement to satellites and provides high-resolution, on-demand images using drones.

Disadvantages: the resources of UAVs are scarce. It can also be limited by battery life (usually 15-30 min flight time) to limit the time of operation. The changes in lighting, wind and rain and dust can deteriorate the input of the camera. Make-the-

vision-difficult Motion blur and jitter are caused by overhead motion. In addition, the top-down perspective implies that the man is in the picture as a tiny creature most of the time.

Applicability: The approach is very relevant to search-and-rescue despite being limited as follows. The research always underlines that drones are superior in topography of the disaster area and victimization search. Specifically, the use of signals through gestures may save lives of people in cases, when they could not move (were immobilized) and could not shout at rescuers. Human detection, which can be over the great distance and recognizing local gestures enable a drone to locate individuals, as well as comprehend their requirements e.g. grab their attention by waving.

2.5 Techniques for Human Detection

Aerial human detection is simply object detection applied in a drone image in relation to people. The prevailing paradigm is CNN-based detectors which have been trained on both labelled drone images. Leading method include:

- YOLO Family: The YOLO (You Only Look Once) by Redmon et al. and its variants (YOLO9000, YOLOv2, YOLOv3, YOLOv4, and so on) are in a widespread use. YOLO splits the image into SxS grid and forecasts a constant number of binding packages per cell. It is capable of operating at extremely fast speed (real-time) and more than one person per picture. According to Hossain and Lee, YOLO is usually less accurate than two-stage detectors, but more accurate at classification. YOLOv2 and v3 also have better to capture small objects such as anchor boxes and multi-scale detection. Ideally, on-board inference is performed in numerous UAV papers with YOLOv3-Tiny or YOLOv4- Tiny (a smaller backbone).

2.5.1 Gesture / Pose Detection Methods.

The other thing that needs to be done when the human is discovered is identifying their gesture or pose. Key methods include:

- **Skeleton-Based Recognition:** This method involves two steps, the initial step involves a pose estimator that obtains joint coordinates after which a pose is classified.
- **Gesture recognition:** There are studies that deal with gestures of the hands and not the entire body. Individually, as examples, hand-posture or sign-language recognition on a drone image involve viewpoint invariance, system should recognize the same gesture regardless of whether the individual is facing north or south and scale variations.

2.5.2 Challenges in Real-Time Detection Systems.

UAV vision systems in real time have come across numerous issues:

- **Small, Far Objects:** Aerial imagery frequently its subjects are humans only a couple of dozens of pixels in size, and accuracy is difficult.
- **This includes Fluctuating Weather:** The outdoor images may be heavy in terms of shades, glare, or lack of contrast (dusk or nighttime). Quality also is degraded due to motion blur because of drone movement.

- **Occlusion and Crowds:** Do not get so much of yourself ``abraded by the rubbish or covered by another person in a disaster. The UAV detectors that are currently in use are primarily trained on single subjects. When someone is in the way, or when a crowd is present, it follows that detection and pose-estimation is likely to fail. Last ground scenarios use multi-person pose estimation, whereas the aerial-view joints estimation is less advanced.
- **Compute and Latency Constraints:** It is difficult to show the real-time inference (30 fps or more) on a small drone.
- **Limitations in the Data:** As it was mentioned, the number of large aerial gesture labelled data sets is few.
- **Numerous studies are therefore based on the gathering of data in a lab or model disaster location.**
- **False positives and Safety:** False dust can be risky e.g. the interpretation of a fallen shadow or an animal as somebody waving. The systems should be very precise otherwise; they will destroy the idea of burden rescue teams by producing a lot of noises. On the other hand, a missed person (false negative) may prove to be fatal.be fatal.

2.6 Comparison of drone based human detection vs tradition human detection

Criteria	Drone based	Traditional based
Human detection	Moderate(factors as discussed above)	High
Safety	high	low

Reliability	Low(cant continue long operations)	high
Sensor integration	Many and can be useful	None, nut can be effective for immediate rescue
mobility	Moderate(cannot cover corners and dark areas)	Moderate(can cover, but risks increases)
accuracy	moderate	low

Table 2.1 Drone based human detection vs tradition human detection

2.7 Research Gaps

The literature reviewed has given a number of gaps that stimulate future studies in the area of UAV based disaster rescue:

- **Absence of Outdoor Gesture Data:** The current gesture data of UAVs is almost indoors or controlled. No standard set of emergency rescue signs in the actual disaster situation is available. This lack of information inhibits training and benchmarking.
- **Fixed Gesture Vocabularies:** The existing systems presuppose a limited and stabilized system of gestures. Actual victims may adopt different or culturally-related cues. The existing works do not touch to the full extent on the recognition of unknown or new gestures.
- **Small Object Detection:** As has been mentioned, individuals are extremely tiny in high altitude UAV images.
- **Real-Time Embedded Constraints:** There are limited studies that are concerned with understanding the system-level constraints of real UAV

hardware. Although on-board GPUs are mentioned frequently in the papers, the majority of experiments are performed in laboratory or offline.

3. Problem Statement

When a disaster occurs be it a flood in a city or a landslide in the distant location, when the Traditional rescue teams arrive, they are always faced with a difficulty of finding the survivors within a short time yet the surroundings under such an emergency are too hazardous to be entered in blindly. Although drones have emerged as a major device of aerial surveillant, it has a major weakness. They are like flying cameras and the video is made available to a human operator who has to stare at the screen all night, in hopes of noticing someone waving a hand on a roof or someone buried under rubble. This project aims to improve human safety and efficiency by human detection and gesture interpretation in real-time to deal with such critical situations.

The given project speaks to A human operator who is tired of stress and hours spent at the monitor Port and can miss a 1-second signal of distress. It is a human observation that we would like to substitute with AI-Assisted Drone For Human Detection.

3.1 Methods and Methodology / Approach.

- **Hardware Synchronization:** Pixhawk: The Pixhawk is the flight stabilization and GPS telemetry unit, whereas the Raspberry Pi serves as the brain, which works with video feeds. They interact through the MAVLink protocol so that the drone is made capable of hovering around or manipulating itself to follow discerned visual signals.
- **AI Vision Pipeline:** The system operates with the help of the YOLO-based model to detect humans during the first stage. Engine finds skeletal landmarks to determine the position of hands of a person after he has been detected in relation to shoulders and hips.
- **Temporal Robustness logic:** To avoid the occurrence of false alarms due to accidental movement, a GESTUREHOLLDURATION has been adopted by the system. The Gesture is sensed with 1sec or minimum in such a way that the false alarm can be prevented.
- **Emergency Alerting:** when a distress gesture is confirmed the system captures.
- **Burst Mode** (one image per second), sends them through the Telegram Bot API, spoofed or real GPS position of the rescue team.

3.2 Objectives

Objective No.	Statement of the Objective	Method/ Methodology	Resources Utilised
1	Conduct a comprehensive literature review on human and gesture detection algorithms.	Systematic review of research papers on YOLO, OpenCV, for real-time edge processing.	IEEE Xplore, ResearchGate, Google Scholar.
2	Design and assemble a drone platform integrated with a flight controller and companion computer.	Integration of a Pixhawk 2.4.8 with a Raspberry Pi 5 (16GB) using a MAVLink connection for hardware-software synchronization.	F450 Frame, Pixhawk FC, Raspberry Pi 5, LiPo Battery, 1000kv Motors.
3	Implement realtime image acquisition and processing for dynamic object tracking	Development of a vision pipeline using OpenCV to detect human presence and track movements in real-time.	Arducam imx519 16mp, Python, OpenCV Library.
4	Establish a robust communication and alerting system for emergency response.	Implementation of the Telegram Bot API to send instant photo alerts with metadata	Telegram API, LTE/Wi-Fi Module, MAVLink Protocol.

		(GPS/Altitude) when a distress gesture is detected.	
5	Validate system performance through simulated and real-world flight testing.	Testing "Temporal Robustness" by requiring a 1.0s gesture hold duration to reduce false positives	Mission Planner Software, Open-field test site.

Table 3.1 Objectives

4.Problem Solving

In this chapter, the author describes the process of engineering which was applied to the translation of the theoretical goals of the project into a practical reality. The design process consisted of three important stages, which included Propulsion System Engineering, Avionics Integration, and AI Logic Implementation.

The most critical question that was answered during this step was how to maximize the Thrust-to-Weight Ratio to be able to include the bulky computational load (Raspberry Pi 5) without compromising the flight stability or safety.

4.1 Hardware Architecture and Component Selection.

In order to guarantee a well-built platform and a powerful one that could survive outdoors and be able to process its edges, the following particular components have been chosen and incorporated: integrated:

Component	Component Specification	Motive
Airframe	F450 Quadcopter Frame	High structural integrity with integrated PCB points.
Propulsion	1000KV BLDC Motors (A2212)	Provides high RPM efficiency for 10-inch propellers.
Speed Control	30A SimonK ESCs	Fast refresh rates (400Hz) optimized for multi-rotor stability.

Power Plant	5200mAh 3S LiPo (35C)	High capacity for extended flight times; 35C burst rating ensures safety.
Flight Brain	Pixhawk 2.4.8	Redundant IMU and barometer for precise GPS loiter modes
Companion Computer	Raspberry Pi 5 (8GB/16GB)	YOLOv8,image processing.

Table 4.1 Hardware Architecture and Component Selection

4.2 Engineering Calculations and Propulsion Analysis

A mathematical feasibility study was carried out with ascertaining the possibility of the selection of 1000KV motors used and subsequently 30A ESCs so that the system could be lifted safely before assembly.

Budgets of weight are calculated as a portion of segregated weight per unit (cubic meter).

4.2.1 Weight Budget Estimation (All-Up Weight)

Budgets of weight are determined as a percentage of segregated weight per unit (cubic meter).

To begin with, the Total Take-off Weight has been established by adding the mass component of the item::

Component Category	Original (1.3 kg)	New Estimate (2 kg)	Change
Airframe & Landing Gear	300 g	450 g	+150 g
Propulsion (Motors + ESCs)	320 g	440 g	+120 g
Energy (Battery Pack)	400 g	800 g	+400 g
Primary Logic (Pixhawk/GPS)	120 g	120 g	0 g
Companion Logic (Pi 5 + Fan)	100 g	110 g	+10 g
Sundries (Wiring, PDB, Props)	60 g	80 g	+20 g
Total Take-Off Weight	1300 g	2000 g	+700 g

4.2.2 Thrust-to-Weight Ratio Calculation

Agility and stability are critical factors to a disaster management drone. A drone that is safe needs a minimum of 2:1 Thrust to weight ratio.

Motor Spec: The 1000KV motor is of a standard A2212 type with 1045 propellers that provides about 800g of thrust at 100% throttle.

Total Available Thrust

$$T_{max} = 4 \times 800g = 3200 \text{ grams}$$

Ratio Calculation

$$\text{Ratio} = \frac{T_{max}}{W_{total}} = \frac{3200}{1300} \approx 2.46 : 1$$

Inference: This is an excellent ratio of 2.46: 1. It will result in the drone flying at about 4045% throttle resulting in plenty of remaining power which can be used during fighting strong winds or fast maneuvers during an emergency.

4.2.3 Battery C-Rating and Current Safety Check

We have to make sure that the 30A SimonK ESCs and the 35C Battery will not burn in full load.

Max Current Draught: Each motor at 100 percent throttle has a draught of about 15A.

$$I_{max_load} = 4 \times 15A = 60 \text{ Amps}$$

ESC Capacity:

$$4 \times 30A = 120A \text{ total capacity. Since } 60A < 120A,$$

the SimonK ESCs are certainly clear of the danger zone.

Battery Safety:

$$I_{battery_max} = \text{Capacity (Ah)} \times \text{C-Rating}$$

$$I_{battery_max} = 5.2 \text{ Ah} \times 35 = 182 \text{ Amps}$$

Conclusion: 182A can be provided by the battery, however, at best, the motors will pull up 60A. The electric circuit is also safe.

4.3.4 Theoretical Flight Time Calculation:

$$\text{Time}_{flight} = \frac{\text{Usable Capacity (Ah)}}{\text{Hover Current (A)}} \times 60 \text{ mins}$$

$$\text{Time}_{flight} = \frac{4.16}{17} \times 60 \approx 14.6 \text{ minutes}$$

4.4 Hardware Integration and Wiring Strategy

The power circulation was done in a Power Distribution Board (PDB).

1. PDB Integration: Battery lead (XT60) is also attached to the PDB. The four SimonK ESCs are peeled off and solder on the pads of the PDB to balance the 11.1 V equally.
2. BEC (Battery Elimination Circuit): Because the voltage is 11.1V AC the Pixhawk and Raspberry Pi are unable to read it directly, a 5V/3A BEC is needed to reduce the voltage to the flight controller.
3. Motor Mapping:
 - o Motor 1 and 2 Counter-Clockwise (CCW)
 - o Motor 3 and 4 Clockwise(CW)
 - o Caution: Poor wiring in this case causes the drone to undergo a flip as soon as it takes off.

4.5 AI Vision Pipeline Implementation

The section describes the software logic that will be executed on the Raspberry Pi

5

4.5.1 The Detection Logic Flowchart

The software is in the form of a continuous Loop. The system is programmed so that it helps in filtering out the noise (random movements) and only activates the SOS-like gestures.

System Logic Flow:

1. Start: Welcome to the Camera and Load YOLOv8 Model.
2. Input Capture Frame (640x640 resolution).
3. Process 1 (Detection):
 - o Load YOLO Inference.
 - o Condition: Is Class Person and Confidence Above 0.6?
 - o NO: Return to Start.
 - o YES: Continue to Pose Estimation.
4. Process 2 (Skeleton Extraction):
 - o Get (X, Y) points at RightWrist, LeftWrist, right, and left shoulders.
5. Process 3 (Geometric Logic):
 - o Checking: check whether YWrist is smaller than YShoulder (in images, Y is smaller on your way up).
 - o Condition: Do both wrists go above shoulders?
6. Process 4 (Temporal Filter):
 - o Timer Check: Has it taken more than 1.0 seconds to be this true?
 - o YES: TRIGGER ALERT.
7. Output: Send the Telegram Alert having GPS location.

4.5.2 Why Raspberry Pi 5?

Past versions with Raspberry Pi 4 were not able to sustain 10 FPS (Frames Per Second). Raspberry Pi 5 offers an important increase in performance, and now it is possible to run the quantized YOLO model with 25-30 FPS, which is vital in real-time tracking when flying.

4.6 Communication System (Telegram Bot)

The communication link does not rely on complex ground station software but uses a standard messaging app for universal accessibility.

- API Token: A unique bot token is generated via "BotFather."
- Chat ID: The rescue team's group ID is hardcoded into the Python script.
- Payload: When the "Trigger Alert" function is called, the script executes:

Python def send_alert(image, gps_lat, gps_lon):

```
caption    =    f"SOS  DETECTED!\nLat:    {gps_lat}\nLon:
    {gps_lon}\nAlt:    15m"    bot.send_photo(chat_id, photo=image,
caption=caption)
```

4.7 Validation Scenarios

To ensure the hardware and software work in unison, the following test bench was established: 1. Bench Test (No Propellers):

- o Power up system.
- o Perform SOS gesture in front of camera.
- o Pass Criteria: Motors should NOT disarm, but the Telegram notification must arrive within 3 seconds.

2. Vibration Test:

- o Run motors at 50% throttle (clamped down).
- o Check for "Jello Effect" in camera feed.
- o Solution: Soft-mount dampers were added under the Pi Camera to reduce rolling shutter distortion caused by the 1000KV motors.

3. Load Test:

- o Hover at 2 meters for 10 minutes.
- o Measure battery voltage and motor temperature.
- o Pass Criteria: Motors should be warm (not hot), and voltage should be $> 10.5V$.

5. Results and Discussions

A natural disaster is a matter of seconds. In this project, the author presents AI-Assisted Drone, which should perform the roles of search and rescue teams. With the integration of the CV and the available equipment, the drone searches the site of the victims and sends real-time notifications to the emergency services in remote locations.

The AI-Assisted Drone For Disaster Management was created by considering, selecting and using basic but effective parts, to come up with a prototype that is capable of identifying humans using the aerial perspective, process the information, and relay it to the telegram API BOT.

5.1 Drone Prototype

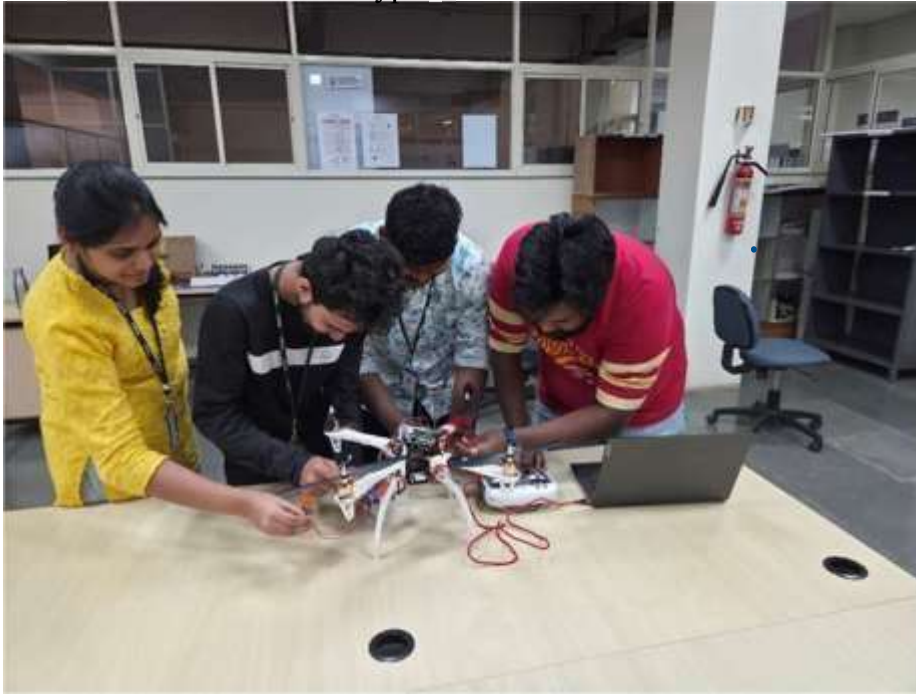


Figure 5.1 Drone Prototype



5.1.2 Drone Arming



Figure 5.2 Drone Arming

5.1.3 Drone Flying



Figure 5.3 Drone flying

5.2 human detection/software integration



Figure 5.4 MAVlink Protocol





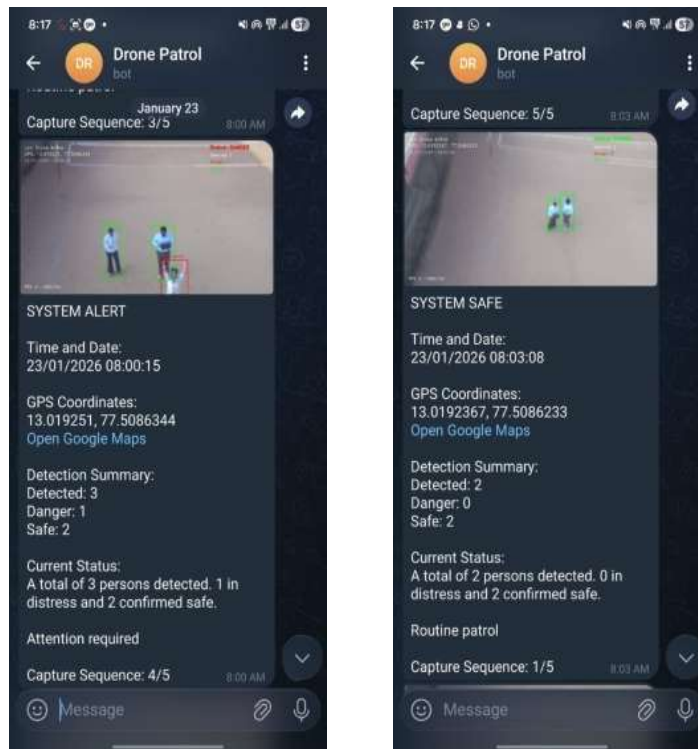


FIG: 5.5 OUT PUT

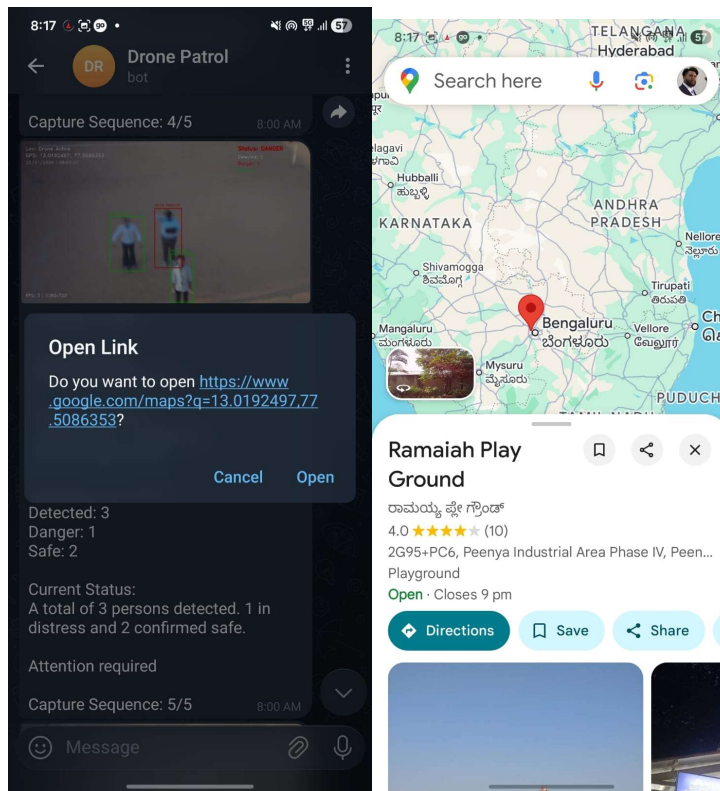


FIG: 5.5 OUT PUT

6. Conclusions and Future Directions

Conclusions:

AI - Assisted Drone for Disaster Management, which is a prototype, is built around a Raspberry pi 5, That is connected to the PixHawk via serial, following a mavlink protocol, whereby on receiving the processed data in rpi, the Mavlink is an post-box and receives one serial stream over UDP to ground station(laptop). Open cv traps the identified frame with yolov8 pose model and it moves the frame to be better visualized by annotating it, it is

then transmitted to a telegram bot API via a connection with the current port onto a computer.

Suggestions for future directions:

- In cases of disaster the flow of communication might be disjointed, hence we must find a method of ensuring that this system has a smooth connectivity.
- Additional camera lens to obtain various forms of captured outputs that are potentially valuable, example: - thermal camera to capture the hot spots of a human being in danger.
- Drone autonomous navigation system, whereby it is able to cover the areas of disasters on its own.

Ambreeesh P

AI - Assisted Drone for Disaster Management, which is a prototype, is built around a Raspberry pi 5

 PAPER ECE_2022 M. S. Ramaiah University Of Applied Sciences

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Abstract

Disasters Floods, earthquakes, fires, and industrial accidents are some of the disasters that often provide unsafe and inaccessible conditions in which quick rescue is difficult. Under this case, it is vital to detect the affected individuals early enough to minimize loss of life. This paper dwells on the Design and Development of an AI-Assisted Drone System to help manage disasters in the natural situation.

The suggested system is interpreted as a quadcopter platform with Pixhawk flight controller board and processing unit on the platform based on Raspberry Pi. It has a camera on the drone capturing real-time video and photographs on the flight, which is analyzed in real-time by AI and CV methods. OpenCV is applied to perform image processing and annotations, whereas human detection is conducted with the help of the YOLOv8 model. The system can detect human presence and recognize any distressed related gesture with the help of a pose estimation model, like a raised hand, and it would need urgent help.

Raspberry PI communicates with the flight controller to maintain a stable flight and handles data properly, the system was able in BOT API to recognize people and produce the appropriate alerts in a timely manner and provide visual information as well as GPS positioning in the case of monitoring when on the ground.

This project points out the idea of the AI-powered drones as the future of disaster response to a better situational awareness, a decrease in human risk, and a quicker nature of the rescue response.

1. Introduction

Every year natural and manmade catastrophes like floods, cyclones, earthquakes, disasters, and fires and industrial mishaps take place in Recent times. The consequences of these disasters result in huge loss of human life and property. In such cases the operations of rescuing will also be challenging due to the geographical areas and can be hazardous to Find rescued individuals in disaster afflicted places. The safety and a delay, which may lead to the human life risk, is one of the greatest issues faced by rescue teams.

Drones have found their use in disaster management since they are able to travel to hazardous and inaccessible locations and give real-time images and videos. This assists authorities to know more about the situation. Nevertheless, drone systems used by majority of people need constant human supervision and this is time consuming and can cause errors.

To counter this drawback, smart systems are necessary that will be able to process visual information by automatic means and implement the Challenges, the development and design of an AI Assisted Drone For Disaster Management is a prototype that will be deployed to help save lives through more efficient identification of issues of distress and human beings.



Figure1.1 Detection of a Distress Person

Design and Development of AI – Assisted Drone for Disaster Management

1

1.1 Visual Challenges in Disaster Scenarios

Disaster The conditions of a disaster are normally unpredictable. There are poor conditions of lighting thus making the analysis hard to visualize. The images taken in case of disasters can be blurred because of the movements of the drones or because of low visibility and then they are surrounded by water and therefore can be detected by human beings with more difficulty. Such poor-quality images are one of the primary research issues which involve identifying humans.

Under such circumstances, image processing approaches are liable to failure.

Image recognition systems should be in a position of dealing with noise, and varying lighting conditions.

One of the significant research issues in the field of disaster management application is the development of such robust systems.

This project examines that image recognition methods using AI can be used to identify humans even on challenging environments when they are considered

difficult to view. Through concentration on real time image analysis using Visual Enhancement techniques the system will enhance reliability of the human detection in times of disaster.

The image-based monitoring technique is very important in detecting the presence of drones near airports and within runways.

1.2 Significance of Drones in Image-Based Monitoring

Image-based monitoring method is quite essential in identifying the existence of drones in around airports and runways.

In the event of a disaster, the use of drone is of great significance to the present-day disaster monitoring. They offer aerial images which cannot be viewed through the ground-based techniques. A drone has the capability to serve a large area in a limited time period and give real time images of the point of disaster, This assists the authorities to have visuals of the damages and organize rescue missions more efficiently. This data is then too hard to analyse manually and is life threatening. This brings out the necessity of automated systems of monitoring through images. Together with image recognition methods, it is also possible to develop systems capable of capturing images and, at the same time, analyse them intelligently. This project aims at enhancing the practicality of monitoring using drones through image recognition.

1.3 Motivation and Need for the project

long rescue operations usually arranged by local parties or NDMA, usually put their lives in danger when undertaking rescue operations,

Human operators need to pay full attention to the video feeds of the drones and this makes them fatigued and they can overlook critical information.

Automatically performing an analysis of images and recognizing humans is done by image recognition techniques. With AI, drones can identify individuals on the spot,

and emphasize on valuable information on sight. This minimizes manpower required by the human operators and increases the speed of response.

In the project, humans can be detected by their image recognition, and basic gestures that can also report distress are identified. The system takes the live video frames and gives useful information to the rescue teams. This proves that image recognition is significant in enhancing disaster response mechanisms.

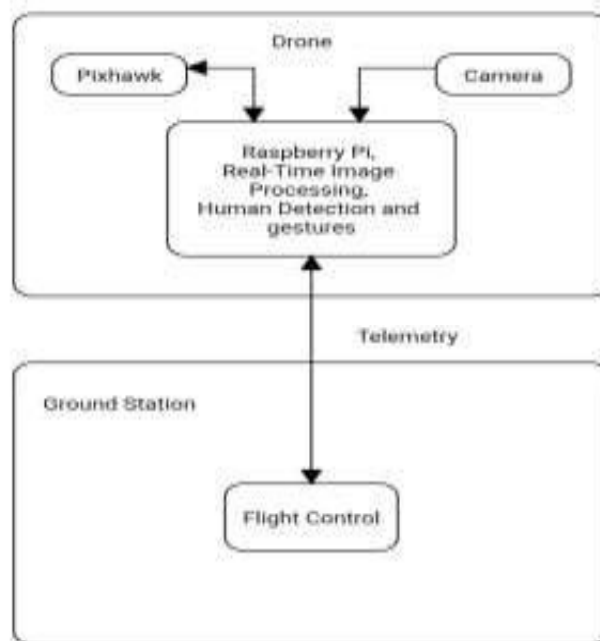


Figure 1.2 Block diagram of the system

1.4 Future prospects and potential enhancements

In a state such as India, where natural calamities are common, and they affect huge numbers of people, vision bio type systems may be crucial in enhancing effectiveness and safety during rescue operations. To some cases the communication aspects might be compromised, so the communication of the drone to the ground station should be more

improved and enhanced, The likes of fish camera lens and thermal cameras among other camera lens might be used to see a better result of the situation i.e. a person trapped under a slab or in a building the use of thermal cameras might help.

2. Background Theory

By Through the study of the previous work, mapping of how similar problems have been approached in the past and the methods used, visual analysis and the research of the limitations of the established systems is set, like the manual detection of a human body will occupy lots of labour and time wasting which in such situations is very essential and such research has high percentile of error.

Various researches have focused on UAV based detections, including a basic examination of conditions to self-flying drone, although in most cases there is the challenge of drone control, image enhancements and communications, low range and certain challenges few.

2.1 Foundational Studies in UAV-based Human Detection.

Some of the early approaches to human monitoring with the use of UAVs relied on more traditional computer vision protocols, Gradients were coined with established classifiers such as

Local Binary Patterns and FPDW Support Vector Machines to detect humans in aerial images Combination of measures such as HOG with SVM, Local Binary Patterns and FPDW had fair robustness to variation, though their performance critically relied on the selection and tuning of features.

At the same time, there were drones used in photography and general surveillance. Later experiments commenced on more specific uses in the context of human detection and interaction. it is explicitly indicated in these studies that the drones with cameras can observe and analyse human presence in aerial perspectives. Nevertheless, the majority of these early works were aimed only at human detection as object and did not regard the analysis of the body posture or meaningful gesture acknowledgements.

2.2 Principles and Assumptions

The aerial human gesture detection systems rely on a number of geometric as well as operational assumptions. In other words, the size of the human beings in the image is defined by the altitude of the drone and the resolution of its camera. Practically, the bigger the resolution of the camera, the higher the flight altitude, provided that we indulge too much in high flight or have a low-resolution camera, the pixels are small which is detrimental to detecting human beings.

Algorithms.

On the algorithmic side, a large number of models are based on the ability to use large training datasets of labelled examples. In reality, aerial vision systems cannot work without people of interest being seen either obliquely or above, and without the model having been trained or fined-tuned on visually similar images.

The major assumptions are:

- Drones' stability: it should be as stable as possible so that the camera does not get blurry products, its stability depends on several factors such as Central Gravity(CG) and aerodynamic disturbance.
- loiter mode: the drone should be stationary when an operation is in process, like the period of time the data-analytics may be perfect and it may be utilized in mapping an rescue strategy.
- environmental conditions: the drone should not be affected by environmental factors like the wind, turbulence etc.
- low light improvement method: the low light situation renders the image of the human being very hazy and unattainable thus improving the +video is an added advantage in the given project.
- High resolution: makes the detection from a far visual point easy, as it can clearly detect and analyse the key points in the captured frames.

2.3Key Definitions

- UAV (drone): This is an unmanned flying vehicle which has cameras/sensors.
- Object Detection: The object recognition and classification problem in an image (persons in this case).
- Human Pose Estimation: Detecting important joints locations (e.g. elbows, knees) of a person. Usually in the form of a 2D skeleton or heatmaps.
- Gesture Recognition: Categorizing a human body pose or movement into a redefined set Gestures here are restricted to a set of rescue gestures such as waving arms to draw attention.
- Real-Time System: This is a system that can autonomously handle every video frame (or a video stream) at a sufficiently fast frame rate (typically [?]30 fps) to be interactive.

- **System Throughput:** The frames per second (FPS) value that is obtained points to viability in real time. On board UAV systems are usually aimed at 10- 30 FPS based on equipment.
- **Drone stability:** Stability in this context entails the fact that despite the effects on the environment and regardless of the weakness in the structural parts the drone remains at a stable position.
- **loiter mode:** Loiter Mode is an automatic mode, and it tries to keep the current location, heading and altitude.
- **environmental conditions** There are a few major conditions that influence drone surveying, which include temperature, wind, humidity, rain, and clouds.
- **image enhancement on low light:** low-light image improving process is capable of raising image brightness and more accurately restoring colour and detail information.
- **High quality:** High quality drones refer to those drones that have the capacity to take detailed imagery, which are required in professional aerial photographing, inspection, and surveying.

2.4 Merits, Demerits and Applicability.

Merits: Aerial vision gives various situational awarenesses. Drones are able to reach isolated or unsafe areas and survey a greater area in a short period and provide an aerial perspective unattainable to ground troops. They are able to travel tens of square kilometres per hour, and thus can search victims fast. Detection based on vision is non-invasive: the wearable sensors do not make the victim carry an item unlike vision-based detection. Shopping Cart 2 is also a supplement to satellites and provides high-resolution, on-demand images using drones.

Disadvantages: the resources of UAVs are scarce. It can also be limited by battery life (usually 15-30 min flight time) to limit the time of operation. The changes in lighting, wind and rain and dust can deteriorate the input of the camera. Make-the-

vision-difficult Motion blur and jitter are caused by overhead motion. In addition, the top-down perspective implies that the man is in the picture as a tiny creature most of the time.

Applicability: The approach is very relevant to search-and-rescue despite being limited as follows. The research always underlines that drones are superior in topography of the disaster area and victimization search. Specifically, the use of signals through gestures may save lives of people in cases, when they could not move (were immobilized) and could not shout at rescuers. Human detection, which can be over the great distance and recognizing local gestures enable a drone to locate individuals, as well as comprehend their requirements e.g. grab their attention by waving.

2.5 Techniques for Human Detection

Aerial human detection is simply object detection applied in a drone image in relation to people. The prevailing paradigm is CNN-based detectors which have been trained on both labelled drone images. Leading method include:

- YOLO Family: The YOLO (You Only Look Once) by Redmon et al. and its variants (YOLO9000, YOLOv2, YOLOv3, YOLOv4, and so on) are in a widespread use. YOLO splits the image into SxS grid and forecasts a constant number of binding packages per cell. It is capable of operating at extremely fast speed (real-time) and more than one person per picture. According to Hossain and Lee, YOLO is usually less accurate than two-stage detectors, but more accurate at classification. YOLOv2 and v3 also have better to capture small objects such as anchor boxes and multi-scale detection. Ideally, on-board inference is performed in numerous UAV papers with YOLOv3-Tiny or YOLOv4- Tiny (a smaller backbone).

2.5.1 Gesture / Pose Detection Methods.

The other thing that needs to be done when the human is discovered is identifying their gesture or pose. Key methods include:

- **Skeleton-Based Recognition:** This method involves two steps, the initial step involves a pose estimator that obtains joint coordinates after which a pose is classified.
- **Gesture recognition:** There are studies that deal with gestures of the hands and not the entire body. Individually, as examples, hand-posture or sign-language recognition on a drone image involve viewpoint invariance, system should recognize the same gesture regardless of whether the individual is facing north or south and scale variations.

2.5.2 Challenges in Real-Time Detection Systems.

UAV vision systems in real time have come across numerous issues:

- **Small, Far Objects:** Aerial imagery frequently its subjects are humans only a couple of dozens of pixels in size, and accuracy is difficult.
- **This includes Fluctuating Weather:** The outdoor images may be heavy in terms of shades, glare, or lack of contrast (dusk or nighttime). Quality also is degraded due to motion blur because of drone movement.

- **Occlusion and Crowds:** Do not get so much of yourself ``abraded by the rubbish or covered by another person in a disaster. The UAV detectors that are currently in use are primarily trained on single subjects. When someone is in the way, or when a crowd is present, it follows that detection and pose-estimation is likely to fail. Last ground scenarios use multi-person pose estimation, whereas the aerial-view joints estimation is less advanced.
- **Compute and Latency Constraints:** It is difficult to show the real-time inference (30 fps or more) on a small drone.
- **Limitations in the Data:** As it was mentioned, the number of large aerial gesture labelled data sets is few.
- **Numerous studies are therefore based on the gathering of data in a lab or model disaster location.**
- **False positives and Safety:** False dust can be risky e.g. the interpretation of a fallen shadow or an animal as somebody waving. The systems should be very precise otherwise; they will destroy the idea of burden rescue teams by producing a lot of noises. On the other hand, a missed person (false negative) may prove to be fatal.be fatal.

2.6 Comparison of drone based human detection vs tradition human detection

Criteria	Drone based	Traditional based
Human detection	Moderate(factors as discussed above)	High
Safety	high	low

Reliability	Low(cant continue long operations)	high
Sensor integration	Many and can be useful	None, nut can be effective for immediate rescue
mobility	Moderate(cannot cover corners and dark areas)	Moderate(can cover, but risks increases)
accuracy	moderate	low

Table 2.1 Drone based human detection vs tradition human detection

2.7 Research Gaps

The literature reviewed has given a number of gaps that stimulate future studies in the area of UAV based disaster rescue:

- **Absence of Outdoor Gesture Data:** The current gesture data of UAVs is almost indoors or controlled. No standard set of emergency rescue signs in the actual disaster situation is available. This lack of information inhibits training and benchmarking.
- **Fixed Gesture Vocabularies:** The existing systems presuppose a limited and stabilized system of gestures. Actual victims may adopt different or culturally-related cues. The existing works do not touch to the full extent on the recognition of unknown or new gestures.
- **Small Object Detection:** As has been mentioned, individuals are extremely tiny in high altitude UAV images.
- **Real-Time Embedded Constraints:** There are limited studies that are concerned with understanding the system-level constraints of real UAV

hardware. Although on-board GPUs are mentioned frequently in the papers, the majority of experiments are performed in laboratory or offline.

3. Problem Statement

When a disaster occurs be it a flood in a city or a landslide in the distant location, when the Traditional rescue teams arrive, they are always faced with a difficulty of finding the survivors within a short time yet the surroundings under such an emergency are too hazardous to be entered in blindly. Although drones have emerged as a major device of aerial surveillant, it has a major weakness. They are like flying cameras and the video is made available to a human operator who has to stare at the screen all night, in hopes of noticing someone waving a hand on a roof or someone buried under rubble. This project aims to improve human safety and efficiency by human detection and gesture interpretation in real-time to deal with such critical situations.

The given project speaks to A human operator who is tired of stress and hours spent at the monitor Port and can miss a 1-second signal of distress. It is a human observation that we would like to substitute with AI-Assisted Drone For Human Detection.

3.1 Methods and Methodology / Approach.

- **Hardware Synchronization:** Pixhawk: The Pixhawk is the flight stabilization and GPS telemetry unit, whereas the Raspberry Pi serves as the brain, which works with video feeds. They interact through the MAVLink protocol so that the drone is made capable of hovering around or manipulating itself to follow discerned visual signals.
- **AI Vision Pipeline:** The system operates with the help of the YOLO-based model to detect humans during the first stage. Engine finds skeletal landmarks to determine the position of hands of a person after he has been detected in relation to shoulders and hips.
- **Temporal Robustness logic:** To avoid the occurrence of false alarms due to accidental movement, a GESTUREHOLLDURATION has been adopted by the system. The Gesture is sensed with 1sec or minimum in such a way that the false alarm can be prevented.
- **Emergency Alerting:** when a distress gesture is confirmed the system captures.
- **Burst Mode** (one image per second), sends them through the Telegram Bot API, spoofed or real GPS position of the rescue team.

3.2 Objectives

Objective No.	Statement of the Objective	Method/ Methodology	Resources Utilised
1	Conduct a comprehensive literature review on human and gesture detection algorithms.	Systematic review of research papers on YOLO, OpenCV, for real-time edge processing.	IEEE Xplore, ResearchGate, Google Scholar.
2	Design and assemble a drone platform integrated with a flight controller and companion computer.	Integration of a Pixhawk 2.4.8 with a Raspberry Pi 5 (16GB) using a MAVLink connection for hardware-software synchronization.	F450 Frame, Pixhawk FC, Raspberry Pi 5, LiPo Battery, 1000kv Motors.
3	Implement realtime image acquisition and processing for dynamic object tracking	Development of a vision pipeline using OpenCV to detect human presence and track movements in real-time.	Arducam imx519 16mp, Python, OpenCV Library.
4	Establish a robust communication and alerting system for emergency response.	Implementation of the Telegram Bot API to send instant photo alerts with metadata	Telegram API, LTE/Wi-Fi Module, MAVLink Protocol.

		(GPS/Altitude) when a distress gesture is detected.	
5	Validate system performance through simulated and real-world flight testing.	Testing "Temporal Robustness" by requiring a 1.0s gesture hold duration to reduce false positives	Mission Planner Software, Open-field test site.

Table 3.1 Objectives

4.Problem Solving

In this chapter, the author describes the process of engineering which was applied to the translation of the theoretical goals of the project into a practical reality. The design process consisted of three important stages, which included Propulsion System Engineering, Avionics Integration, and AI Logic Implementation.

The most critical question that was answered during this step was how to maximize the Thrust-to-Weight Ratio to be able to include the bulky computational load (Raspberry Pi 5) without compromising the flight stability or safety.

4.1 Hardware Architecture and Component Selection.

In order to guarantee a well-built platform and a powerful one that could survive outdoors and be able to process its edges, the following particular components have been chosen and incorporated: integrated:

Component	Component Specification	Motive
Airframe	F450 Quadcopter Frame	High structural integrity with integrated PCB points.
Propulsion	1000KV BLDC Motors (A2212)	Provides high RPM efficiency for 10-inch propellers.
Speed Control	30A SimonK ESCs	Fast refresh rates (400Hz) optimized for multi-rotor stability.

Power Plant	5200mAh 3S LiPo (35C)	High capacity for extended flight times; 35C burst rating ensures safety.
Flight Brain	Pixhawk 2.4.8	Redundant IMU and barometer for precise GPS loiter modes
Companion Computer	Raspberry Pi 5 (8GB/16GB)	YOLOv8,image processing.

Table 4.1 Hardware Architecture and Component Selection

4.2 Engineering Calculations and Propulsion Analysis

A mathematical feasibility study was carried out with ascertaining the possibility of the selection of 1000KV motors used and subsequently 30A ESCs so that the system could be lifted safely before assembly.

Budgets of weight are calculated as a portion of segregated weight per unit (cubic meter).

4.2.1 Weight Budget Estimation (All-Up Weight)

Budgets of weight are determined as a percentage of segregated weight per unit (cubic meter).

To begin with, the Total Take-off Weight has been established by adding the mass component of the item::

Component Category	Original (1.3 kg)	New Estimate (2 kg)	Change
Airframe & Landing Gear	300 g	450 g	+150 g
Propulsion (Motors + ESCs)	320 g	440 g	+120 g
Energy (Battery Pack)	400 g	800 g	+400 g
Primary Logic (Pixhawk/GPS)	120 g	120 g	0 g
Companion Logic (Pi 5 + Fan)	100 g	110 g	+10 g
Sundries (Wiring, PDB, Props)	60 g	80 g	+20 g
Total Take-Off Weight	1300 g	2000 g	+700 g

4.2.2 Thrust-to-Weight Ratio Calculation

Agility and stability are critical factors to a disaster management drone. A drone that is safe needs a minimum of 2:1 Thrust to weight ratio.

Motor Spec: The 1000KV motor is of a standard A2212 type with 1045 propellers that provides about 800g of thrust at 100% throttle.

Total Available Thrust

$$T_{max} = 4 \times 800g = 3200 \text{ grams}$$

Ratio Calculation

$$\text{Ratio} = \frac{T_{max}}{W_{total}} = \frac{3200}{1300} \approx 2.46 : 1$$

Inference: This is an excellent ratio of 2.46: 1. It will result in the drone flying at about 4045% throttle resulting in plenty of remaining power which can be used during fighting strong winds or fast maneuvers during an emergency.

4.2.3 Battery C-Rating and Current Safety Check

We have to make sure that the 30A SimonK ESCs and the 35C Battery will not burn in full load.

Max Current Draught: Each motor at 100 percent throttle has a draught of about 15A.

$$I_{max_load} = 4 \times 15A = 60 \text{ Amps}$$

ESC Capacity:

$$4 \times 30A = 120A \text{ total capacity. Since } 60A < 120A,$$

the SimonK ESCs are certainly clear of the danger zone.

Battery Safety:

$$I_{battery_max} = \text{Capacity (Ah)} \times \text{C-Rating}$$

$$I_{battery_max} = 5.2 \text{ Ah} \times 35 = 182 \text{ Amps}$$

Conclusion: 182A can be provided by the battery, however, at best, the motors will pull up 60A. The electric circuit is also safe.

4.3.4 Theoretical Flight Time Calculation:

$$\text{Time}_{flight} = \frac{\text{Usable Capacity (Ah)}}{\text{Hover Current (A)}} \times 60 \text{ mins}$$

$$\text{Time}_{flight} = \frac{4.16}{17} \times 60 \approx 14.6 \text{ minutes}$$

4.4 Hardware Integration and Wiring Strategy

The power circulation was done in a Power Distribution Board (PDB).

1. PDB Integration: Battery lead (XT60) is also attached to the PDB. The four SimonK ESCs are peeled off and solder on the pads of the PDB to balance the 11.1 V equally.
2. BEC (Battery Elimination Circuit): Because the voltage is 11.1V AC the Pixhawk and Raspberry Pi are unable to read it directly, a 5V/3A BEC is needed to reduce the voltage to the flight controller.
3. Motor Mapping:
 - o Motor 1 and 2 Counter-Clockwise (CCW)
 - o Motor 3 and 4 Clockwise(CW)
 - o Caution: Poor wiring in this case causes the drone to undergo a flip as soon as it takes off.

4.5 AI Vision Pipeline Implementation

The section describes the software logic that will be executed on the Raspberry Pi

5

4.5.1 The Detection Logic Flowchart

The software is in the form of a continuous Loop. The system is programmed so that it helps in filtering out the noise (random movements) and only activates the SOS-like gestures.

System Logic Flow:

1. Start: Welcome to the Camera and Load YOLOv8 Model.
2. Input Capture Frame (640x640 resolution).
3. Process 1 (Detection):
 - o Load YOLO Inference.
 - o Condition: Is Class Person and Confidence Above 0.6?
 - o NO: Return to Start.
 - o YES: Continue to Pose Estimation.
4. Process 2 (Skeleton Extraction):
 - o Get (X, Y) points at RightWrist, LeftWrist, right, and left shoulders.
5. Process 3 (Geometric Logic):
 - o Checking: check whether YWrist is smaller than YShoulder (in images, Y is smaller on your way up).
 - o Condition: Do both wrists go above shoulders?
6. Process 4 (Temporal Filter):
 - o Timer Check: Has it taken more than 1.0 seconds to be this true?
 - o YES: TRIGGER ALERT.
7. Output: Send the Telegram Alert having GPS location.

4.5.2 Why Raspberry Pi 5?

Past versions with Raspberry Pi 4 were not able to sustain 10 FPS (Frames Per Second). Raspberry Pi 5 offers an important increase in performance, and now it is possible to run the quantized YOLO model with 25-30 FPS, which is vital in real-time tracking when flying.

4.6 Communication System (Telegram Bot)

The communication link does not rely on complex ground station software but uses a standard messaging app for universal accessibility.

- API Token: A unique bot token is generated via "BotFather."
- Chat ID: The rescue team's group ID is hardcoded into the Python script.
- Payload: When the "Trigger Alert" function is called, the script executes:

Python def send_alert(image, gps_lat, gps_lon):

```
caption    =    f"SOS  DETECTED!\nLat:    {gps_lat}\nLon:
    {gps_lon}\nAlt:    15m"    bot.send_photo(chat_id, photo=image,
caption=caption)
```

4.7 Validation Scenarios

To ensure the hardware and software work in unison, the following test bench was established: 1. Bench Test (No Propellers):

- o Power up system.
- o Perform SOS gesture in front of camera.
- o Pass Criteria: Motors should NOT disarm, but the Telegram notification must arrive within 3 seconds.

2. Vibration Test:

- o Run motors at 50% throttle (clamped down).
- o Check for "Jello Effect" in camera feed.
- o Solution: Soft-mount dampers were added under the Pi Camera to reduce rolling shutter distortion caused by the 1000KV motors.

3. Load Test:

- o Hover at 2 meters for 10 minutes.
- o Measure battery voltage and motor temperature.
- o Pass Criteria: Motors should be warm (not hot), and voltage should be > 10.5V.

5. Results and Discussions

A natural disaster is a matter of seconds. In this project, the author presents AI-Assisted Drone, which should perform the roles of search and rescue teams. With the integration of the CV and the available equipment, the drone searches the site of the victims and sends real-time notifications to the emergency services in remote locations.

The AI-Assisted Drone For Disaster Management was created by considering, selecting and using basic but effective parts, to come up with a prototype that is capable of identifying humans using the aerial perspective, process the information, and relay it to the telegram API BOT.

5.1 Drone Prototype

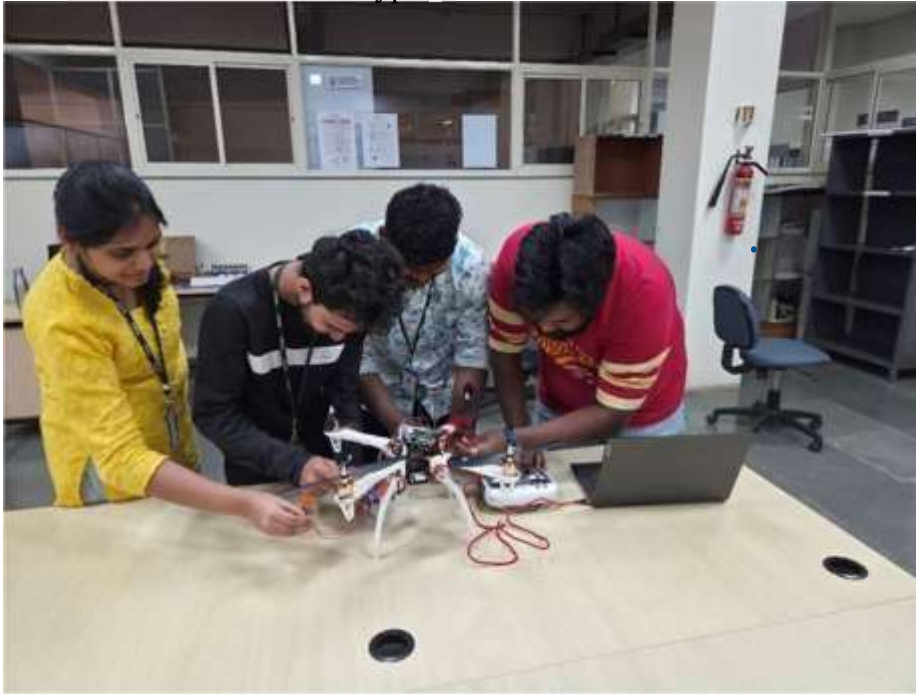


Figure 5.1 Drone Prototype



5.1.2 Drone Arming



Figure 5.2 Drone Arming

5.1.3 Drone Flying



Figure 5.3 Drone flying

5.2 human detection/software integration



Figure 5.4 MAVlink Protocol





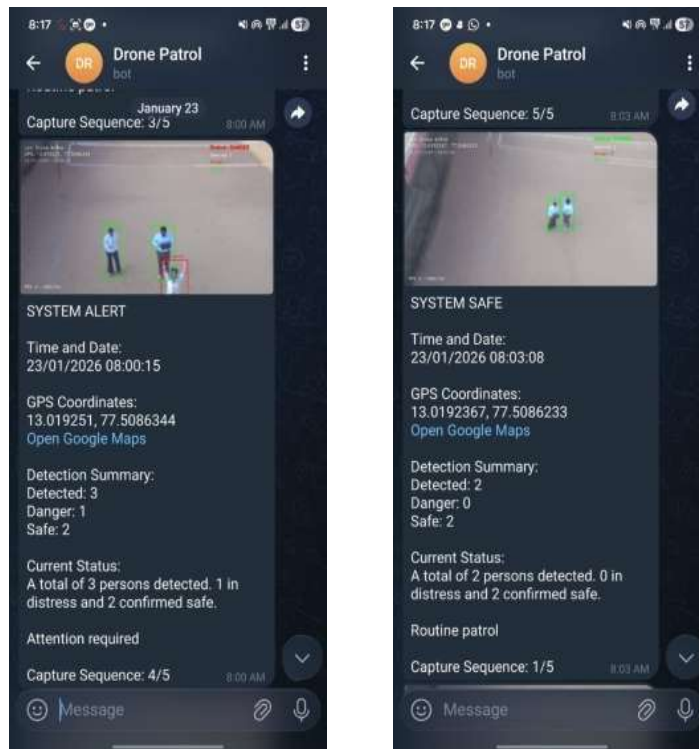


FIG: 5.5 OUT PUT

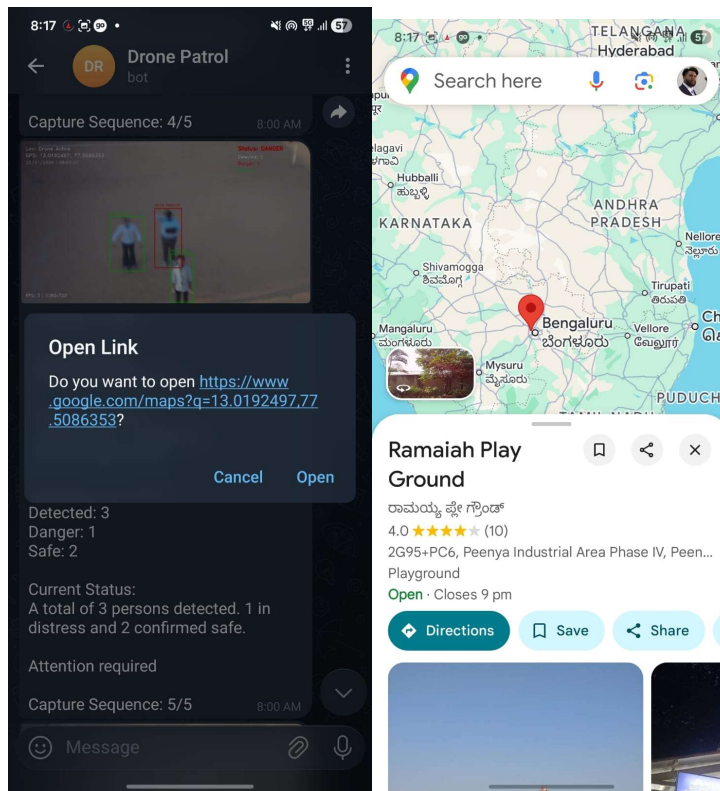


FIG: 5.5 OUT PUT

6. Conclusions and Future Directions

Conclusions:

AI - Assisted Drone for Disaster Management, which is a prototype, is built around a Raspberry pi 5, That is connected to the PixHawk via serial, following a mavlink protocol, whereby on receiving the processed data in rpi, the Mavlink is an post-box and receives one serial stream over UDP to ground station(laptop). Open cv traps the identified frame with yolov8 pose model and it moves the frame to be better visualized by annotating it, it is

then transmitted to a telegram bot API via a connection with the current port onto a computer.

Suggestions for future directions:

- In cases of disaster the flow of communication might be disjointed, hence we must find a method of ensuring that this system has a smooth connectivity.
- Additional camera lens to obtain various forms of captured outputs that are potentially valuable, example: - thermal camera to capture the hot spots of a human being in danger.
- Drone autonomous navigation system, whereby it is able to cover the areas of disasters on its own.